

MAX WAGNER

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Education

Harvard College - A.B. Applied Mathematics/CS + Concurrent Master's • GPA: 3.96/4.0 **Class of 2027**

Relevant Coursework: Stat110: Probability, AM210: Applied Linear Algebra, CS208: Differential Privacy, CS124: Data Structures and Algorithms, AM105: Ordinary and Partial Differential Equations, AM210: Algebraic Fundamentals of Representing Data (Algebraic methods for correlation analysis, causal inference, and ML optimization), AM104: Fourier and Complex Analysis, CS61: Systems (low-level implementations of process control, caching, concurrency, virtual memory). *Spring Coursework:* CS2620: Distributed Computing, CS2050: High Performance Computing (CUDA, OpenMP, MPI, parallelization), AM202: PDEs and their Applications.

Employment

Harvard University SEAS

Undergraduate Researcher, *May 2025-Present*

Solo project - Built a pipeline for diffusion-based single-image-to-3DGS of humans, allowing photos of humans to be transformed into real-time posable and renderable 3D representations, including novel views. Trained Gaussians using diffusion through Pytorch-Lightning on Harvard's cloud cluster through SLURM, optimized view selection with progressive camera angles leading to improved faithfulness to baseline image.

Decipad

Software Engineering Intern, *May 2024-Dec 2024*

Led project to build infrastructure using Rust WebAssembly (WASM) to accelerate big data ingestion and processing for a web-based data notebook product. Constructed a robust and flexible type system in Rust WASM to interface with TypeScript code and seamlessly transfer data and accelerate data processing. Worked with synchronization and lazy updates as well as parallel computing to optimize web app performance. Scaled compute performance by multiple orders of magnitude, allowing 100x-1000x larger datasets to be processed on-device through the web. Integrated work with the typescript team with robust testing and serialization efforts.

Cellino Biotech, Inc.

Engineering Intern, *May - August 2023*

Built image/video processing pipelines and automated fluidic systems to quantify flow profiles through enclosed cell growth chambers in a self-led team of interns. Extensive algorithm development and use of Python for instrument control, accurate data collection, large-scale data processing, and display. Helped integrate our imaging system with automated pipetting setup to collect and process data. Dec 2024: short follow-on project to create a high-rate, hot-swappable, and easy to use cell culture confluence quantifier.

Skills

Programming & AI: Python (Tensorflow, Pytorch, Pytorch-Lightning, data analytics, visualization, image and video processing, optimization, threestudio), Rust (runtime optimization, robust type systems, parallelization, WASM), C++ (concurrency, optimization, low-level control), MATLAB, JMP.

Systems & Tools: Synchronization, Caching, Process control and communication, WebAssembly, Arduino, large-scale data processing, SLURM, cluster and cloud workflows.

Other Activities

Harvard Club Ice Hockey, Co-Chair of Cabot House Committee, former Intramural Chair and Mascot, Harvard Undergraduate Data Analytics Group. Magic the Gathering, Society of Harvard Magicians.